

◆ 1. BERT (Bidirectional Encoder Representations from Transformers)

👉 Best for: Understanding text (context, intent, meaning)

🌐 Real-Time Use Cases

🔍 1. Smart Search (Google-like search)

- Understands user intent instead of just keywords
 - Example:
 - Query: *"Can I take medicine before food?"*
 - BERT understands context of *before food*
-

💬 2. Chatbots (Customer Support)

- Understands user queries accurately
- Improves response relevance

👉 Example:

- Banking chatbot understanding:
"I lost my card yesterday"
-

📊 3. Log Analysis (DevOps / Monitoring)

- Reads logs and identifies issues
- Classifies errors vs warnings

👉 Example:

- Detects patterns in **application logs (ELK / Splunk)**
-

📄 4. Sentiment Analysis

- Used in:
 - Product reviews
 - Social media monitoring

◆ 2. GPT (Generative Pre-trained Transformer)

👉 Best for: Generating text, automation, conversations

🌐 Real-Time Use Cases

🤖 1. AI Assistants (like ChatGPT)

- Answer questions
- Generate documentation
- Help in coding

💻 2. DevOps Automation

- Generate:
 - Terraform scripts
 - Kubernetes YAML
 - CI/CD pipelines

👉 Example:

- “Create Dockerfile for Node.js app”

🔧 3. Incident Response Automation

- Reads alerts
- Suggests fixes
- Generates runbooks

👉 Example:

- “High CPU usage on EC2” → GPT suggests scaling

📄 4. Content Generation

- Blogs, emails, reports
- Documentation

3. LLMs (Large Language Models)

 Best for: End-to-end intelligent systems (combining BERT + GPT capabilities)

Real-Time Use Cases

1. AIOps (AI for IT Operations)

- Combines logs + metrics + traces
- Detects anomalies
- Suggests root cause

 Tools:

- Dynatrace Davis AI
 - Splunk ITSI
-

2. Cloud Cost Optimization (AWS)

- Analyze usage patterns
 - Recommend:
 - Right-sizing EC2
 - Storage optimization
-

3. CI/CD Pipeline Intelligence

- Detect pipeline failures
 - Suggest fixes
 - Auto-remediation
-

4. Knowledge Assistants

- Internal company knowledge bots
- Answer:

- “How to deploy in EKS?”
- “What is our rollback strategy?”

5. Security & Threat Detection

- Analyze logs for anomalies
- Detect suspicious activity

Quick Comparison

Feature	BERT	GPT	LLM
Type	Encoder	Decoder	Combined
Strength	Understanding	Generation	End-to-end intelligence
Example	Search engine	ChatGPT	AI Ops systems

Simple Summary

- **BERT → Understands text (context)**
- **GPT → Generates text (content)**
- **LLM → Does both + takes intelligent actions**

Interview Tip

 If asked in interview:

“Where is BERT used?” → Search & NLP understanding

“Where is GPT used?” → Chatbots & automation

“Where are LLMs used?” → AIOps, DevOps, enterprise AI